RULER What's the Real Context Size of Your Long-Context Language Models (2404.06654v3)

Abstract

Abstract

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> o provide a more comprehensive evaluation of long-context LMs, we create a new synthetic benchmark RULER with flexible configurations for customized sequence length and task complexity.

Customizable sequence length and complexity for the RULER benchmark. Introduce multihop tracing and aggregation. More focus on probing the true context length of the models.

Conclusion

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> We observe common failure modes at large context sizes, including the failure to ignore distractors and ineffective utilization of long context

Same failure modes as described in Visual Haystack. RULER is challenging for models that are top-ranked. What makes RULER challenging?

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> Our analysis further reveals the large potential for improvement on RULER and the benefit of scaling model sizes in achieving better long context capabilities.

Propose the scaling of model sizes.

Tables and figures

Models	Claimed Length	Effective Length	4K	8K	16K	32K	64K	128K	Avg.	wAvg. (inc)	wAvg. (dec)
Llama2 (7B)	4K	-	85.6								
Gemini-1.5-Pro	1M	>128K	96.7	95.8	96.0	95.9	95.9	94.4	95.8	95.5 (1st)	96.1 (1st)
GPT-4	128K	64K	96.6	96.3	95.2	93.2	87.0	81.2	91.6	89.0 (2nd)	94.1 (2nd)
Llama3.1 (70B)	128K	64K	96.5	95.8	95.4	94.8	88.4	66.6	89.6	85.5 (4th)	93.7 (3rd)
Qwen2 (72B)	128K	32K	96.9	96.1	94.9	94.1	79.8	53.7	85.9	79.6 (9th)	92.3 (4th)
Command-R-plus (104B)	128K	32K	95.6	95.2	94.2	92.0	84.3	63.1	87.4	82.7 (7th)	92.1 (5th)
GLM4 (9B)	1M	64K	94.7	92.8	92.1	89.9	86.7	83.1	89.9	88.0 (3rd)	91.7 (6th)
Llama3.1 (8B)	128K	32K	95.5	93.8	91.6	87.4	84.7	77.0	88.3	85.4 (5th)	91.3 (7th)
GradientAI/Llama3 (70B)	1M	16K	95.1	94.4	90.8	85.4	80.9	72.1	86.5	82.6 (8th)	90.3 (8th)
Mixtral-8x22B (39B/141B)	64K	32K	95.6	94.9	93.4	90.9	84.7	31.7	81.9	73.5 (11th)	90.3 (9th)
Yi (34B)	200K	32K	93.3	92.2	91.3	87.5	83.2	77.3	87.5	84.8 (6th)	90.1 (10th)
Phi3-medium (14B)	128K	32K	93.3	93.2	91.1	86.8	78.6	46.1	81.5	74.8 (10th)	88.3 (11th)
Mistral-v0.2 (7B)	32K	16K	93.6	91.2	87.2	75.4	49.0	13.8	68.4	55.6 (13th)	81.2 (12th)
LWM (7B)	1M	<4K	82.3	78.4	73.7	69.1	68.1	65.0	72.8	69.9 (12th)	75.7 (13th)
DBRX (36B/132B)	32K	8K	95.1	93.8	83.6	63.1	2.4	0.0	56.3	38.0 (14th)	74.7 (14th)
Together (7B)	32K	4K	88.2	81.1	69.4	63.0	0.0	0.0	50.3	33.8 (15th)	66.7 (15th)
LongChat (7B)	32K	<4K	84.7	79.9	70.8	59.3	0.0	0.0	49.1	33.1 (16th)	65.2 (16th)
LongAlpaca (13B)	32K	<4K	60.6	57.0	56.6	43.6	0.0	0.0	36.3	24.7 (17th)	47.9 (17th)

A general trend of descreasing scores is observed in the models. The decrease is much more prominent for the open-source models that are not so well-known. Gemini, GPT and Llama also perform well, but suffer a performance drop of 30 percent points as the context length increases.

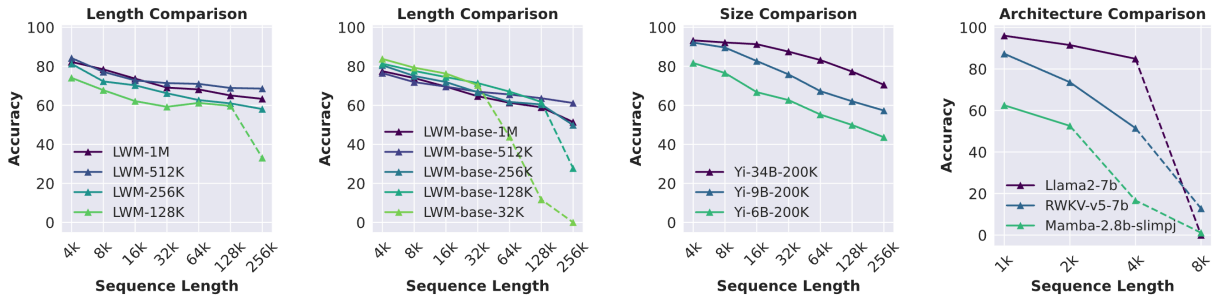

Figure 4: (**Left & middle left**): Comparison of LargeWorldModel (LWM) series trained up to various context sizes with fixed parameter size of 7B. (**Middle right**): Comparison of Yi suite models with different parameter sizes with controlled training context length of 200K. (**Right**): Performance of non-Transformer architectures lags behind the Transformer baseline Llama2-7B by large margin. Length extrapolation is presented with dashed lines.

Introduction

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> synthetic tasks, such as passkey retrieval (Mohtashami & Jaggi, 2023) and needle-in-ahaystack (Kamradt, 2023) to evaluate long-context LMs. However, these evaluations are used inconsistently across works and reveal merely the retrieval capability, failing to gauge other forms of long-context understanding

Synthetic tasks like the ones mentioned abobe test only the retrieval capability and not other forms of long-context understanding. RULER has the following task categories:

1. Retrieval (NIAH)
2. Multi-hop tracing: coreference chain resolution (tracing entities with multi-hop connections)
3. Aggregation: aggregate relevant information
4. QA

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> While all models claim context size of 32k tokens or greater, our results indicate that only half of them can effectively handle sequence length of 32K by exceeding a qualitative threshold.

Evaluation on 18 models, only half of them perform well at a sequence length of 32K.

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> Moreover, almost all models fall below the threshold before reaching the claimed context lengths.

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> urthermore, we observe two behaviors emerging with the scaling of context size across multiple models: the increased reliance on parametric knowledge and the increased tendency to copy from context for non-retrieval tasks

Two important trends

Results and discussion

Benchmark construction

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> Within a constrained domain as in RULER, the task complexity can be thought of as a function of the number of target output tokens and the signal-to-noise ratio in the context.

Retrieval (NIAH)

- Single NIAH: Standard retrieval with single needle. Key value pairs are used
- Multi-keys NIAH: Multiple needles inserted into the haystack. Only one of them needs to be retrieved.
- Multi-values NIAH: Multiple needles with same key. All values associated with the key need to be retrieved
- Multi-query NIAH: Multiple needles with distinct keys. All queried key-value pairs need to be retrieved.

NOTE: RULER also operated in the text modality

Variable tracking

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> Specifically, a variable X1 is initialized with a value V, followed by a linear chain of variable name binding statements (e.g., X2 = X1, X3 = X2, ...), which are inserted at various positions of the input. The objective is to return all variable names pointing to the same value V.

Chaining of variables. Can we do a possible implementation of this in our project?

Aggregation

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> In RULER, we introduce a new category as a proxy for summarization tasks where relevant information constitutes much larger portion of the context, and the target output depends on accurate aggregation of the relevant input. Concretely, we

> construct an input sequence by sampling words from a pre-defined (synthetic) word list.

Question Answering

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> In this task category, we insert the golden paragraphs (i.e., the paragraphs that contain answers) into paragraphs randomly sampled from the same dataset.

This is very similar to our text-to-text setting.

Results

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> Our main goal is to see whether models can maintain such good performance with the scaling of context length. For each task, we evaluate each model with 500 examples generated for each length from the series (4K, 8K, 16K, 32K, 64K, 128K), while complying with each model's necessary chat template.

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> To prevent the model from refusing to answer a query or generating explanations, we append the task input with an answer prefix and check the presence of the target output with recall-based accuracy.

Important configuration detail

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> We notice large performance degradation in all models as we increase input length in RULER. To determine the maximum context size a model can effectively handle, we grade each model with a fixed threshold, passing which indicates satisfactory performance at the length of evaluation. We use the performance of Llama2-7b

> model at the 4K context length as the threshold. We report in Table 3 the maximum length exceeding the threshold as the “effective length” along with the “claimed length”.

Performance above the threshold determined by Llama2-7B is considered "pass" for that context length

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> While the threshold-based grading reveals the discrepancy between claimed and effective length, it lacks details for fine-grained model comparisons. As such, we use a weighted average score to aggregate model performance across various context sizes. We rank models under two weighting schemes: wAvg. (inc) and wAvg. (dec) where the weight linearly increases and decreases with sequence length respectively. Ideally, the weight for each length should be determined by the length distribution of model usage, here we choose the two schemes to simulate the scenarios where longer sequences (inc) or shorter sequences (dec) dominate the distribution.

Scoring scheme

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> he closed-source model Gemini-1.5-Pro outperforms the rest of the models by a large margin, with the effective length greater than the maximum length we have tested on. Pressure testing this model with harder version of RULER can be interesting to follow up in the future.

We can include Gemini 1.5 Pro in our evaluation.

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> all of them exhibit large degradation in RULER as sequence length increases and they fail to maintain performance above the Llama2-7B baseline at their claimed length.

Model performance degrades as the sequence length increases.

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> RoPE (Xiong et al., 2023). Large training context window is not always necessary for good long context performance – top-ranked open-source models contain both brute-force context scaling (Llama3.1 trained on 128K context length) and inference-time length extrapolation (Qwen2 trained on 32K context length). The less performant models also include those trained on much larger context size (e.g., LWM and GradientAI/Llama3 both on 1M context length). Although LWM achieves a higher rank than Mistral-v0.2 when longer sequences receive larger weight (wAvg. inc) and shows less degradation as the context size increases, it performs worse than Llama2-7B even at 4K. This result suggests a trade-off in evaluation between absolute performance on short sequences and the relative degradation with the scaling of context size. We provide more analysis on the model size and maximum training length in section 6.

Findings

1. Brute force scaling and inference-time length extrapolation produce strong long-context performance. Thus training on long-context is not always necessary.
2. Absolute performance of LWM (performace at a fixed context length) is lesser than Mistral, but the relative degradation in performace of LWM over increasing context lengths is better; as shown below.

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Main takeaways:

- Long-context training is not strictly necessary: Models trained on shorter contexts can achieve strong long-context performance through extrapolation techniques.
- Long-context training does not guarantee superior accuracy: Models trained on 1M tokens can still underperform models trained on much shorter sequences.
- Robustness and accuracy are different: A model can be relatively insensitive to increasing context length while having poor absolute performance.

Task error analysis

The analysis is performed on Yi-34B-200K with increased input lengths (up to 256k)

1. Non-robustness to needle types: Needle types other than words cause degraded performance.
2. Susceptibility to distractors
3. Multi-target retrieval is a problem.
4. Copying from context is observed is some models as the context length increases.
5. Unreliable tracking within context.
6. Models use parametric knowledge (like high frequency works: the, an) at higher context lengths when asked to aggregate information.

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> g of the beginning of the input, likely due to the attention sinks (Xiao et al., 2024b).

Question: Is the anchor an attention sink?

Model analysis

Authors study the following:

1. Effect of training context lengths
2. Effect of model size
3. Effect of architecture.

Limitations

1. **Ruler does not have position control:** leads to lack of fine-grained of phenomenon like positional bias
2. **Limited correlation with realistic tasks:** RULER uses synthetic tasks (e.g., variable tracking and frequent-word extraction) as proxies for real-world long-context tasks. However, their effectiveness in representing realistic language understanding remains unverified.
3. **Lack of short-context evaluation:** RULER primarily evaluates performance degradation as context length increases, without sufficiently assessing performance at shorter context lengths. Models may already perform poorly at 4K tokens, making it difficult to distinguish weak baseline capabilities from long-context limitations.

4. **Lack of prompt robustness evaluation:** RULER does not systematically investigate how changes in prompt formatting or task-specific hyperparameters (e.g., variable-name length or vocabulary size) affect model performance.